AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 11 · STUDENT LESSON

Decision Trees

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

AI-2 Week 14: trace a decision tree at its boundaries.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Fit a root question and leaf predictions from training examples. Try the recall before opening notes.

Your learning target

I can fit a **root** question and **leaf prediction**s from **training** examples.

I can trace a new example through the learned tree.

I can compare a shallow tree with a baseline and state its limits.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

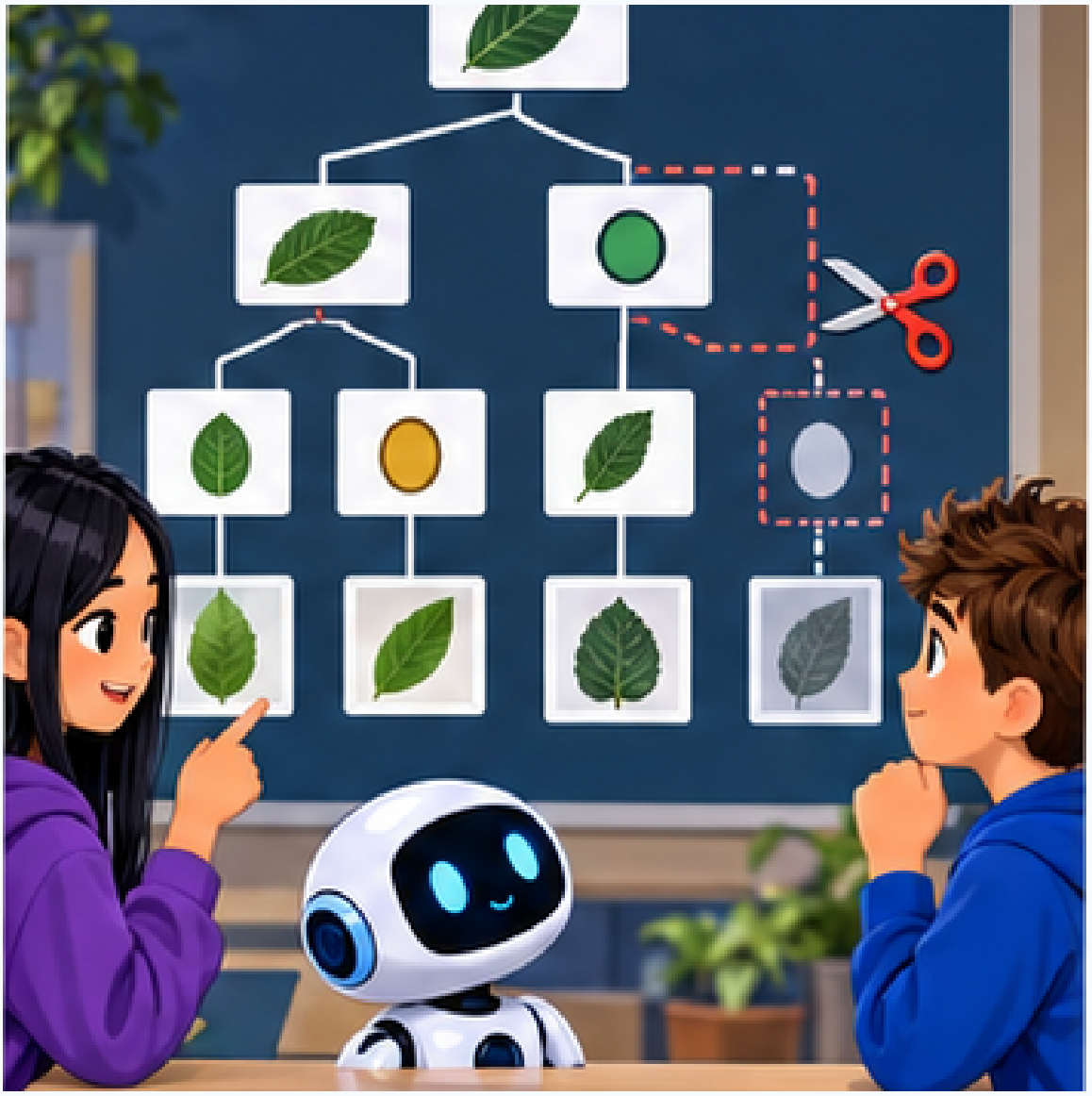
Without opening the old lesson, explain one key idea from Baselines First. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Leakage. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Inspect Before Modeling. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A decision tree partitions feature space using ordered questions; each **leaf** produces a **prediction**.



Your target

LEARNING TARGET Rule: if spots>=4 predict spotted, otherwise healthy. Trace 3,4,5 and missing. Does applying this rule demonstrate that **training** found the **threshold**?

Builder name and experiment date:

Readiness for the connected explanation

Why does a cutoff with zero **training** errors still need validation on separate examples?

This week's end goal

By the end, I can explain decision trees using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

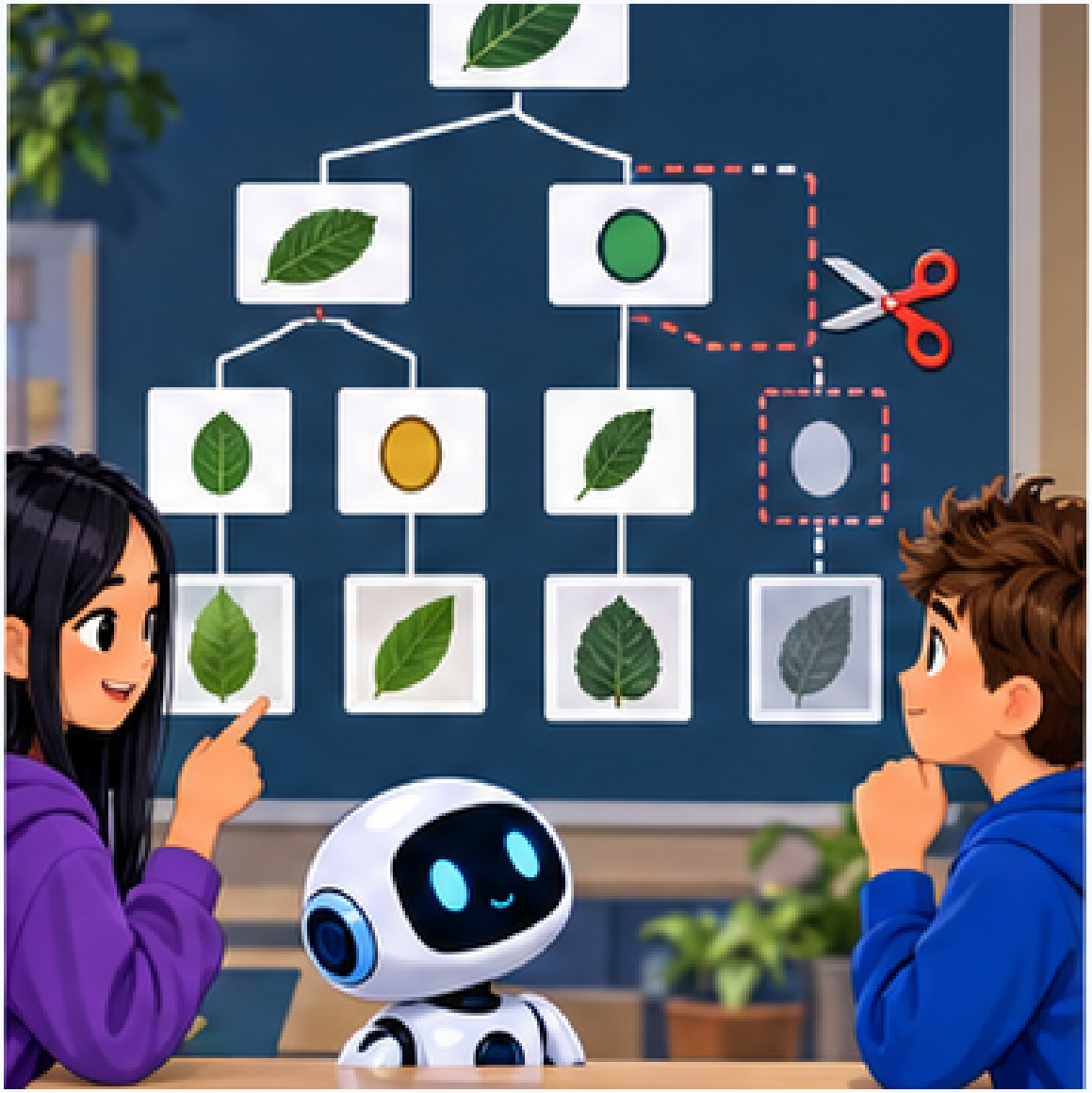
Notice the evidence

Investigate the supplied Decision Trees case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use decision trees to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

A decision tree asks a feature question and follows a **branch** to a **prediction**. A tree with one question has **depth** one and is sometimes called a stump.

Unlike Week 8, where the two **output** labels were fixed in advance, this week we fit both the question and the label stored at each **leaf**.

Our six fictional **training** rows have two binary sensor flags available before **prediction**: near and tall. A value of 1 means that flag is present; 0 means absent. The reviewed scene label is separate.

Each row represents a different trip on a familiar route, including rows with identical values. Similar readings can have different scene labels. These are teaching examples, not robot control instructions.

Row	Near	Tall	Label
1	1	0	blocked
2	1	1	blocked
3	1	0	blocked
4	0	1	clear
5	0	0	clear
6	0	1	blocked

Try near first and tall second. For each question, group the **training** rows by 0 and 1, then predict each group's majority label; choose clear for tied label counts.

Count all **training** mistakes and keep the first question if scores tie. Stop after one question. This is our stated classroom **fitting** procedure.

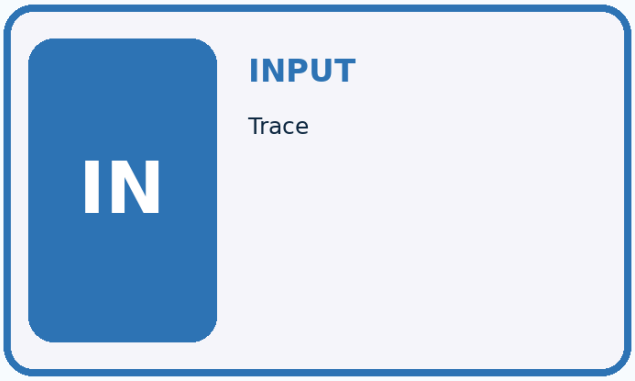
See how it works

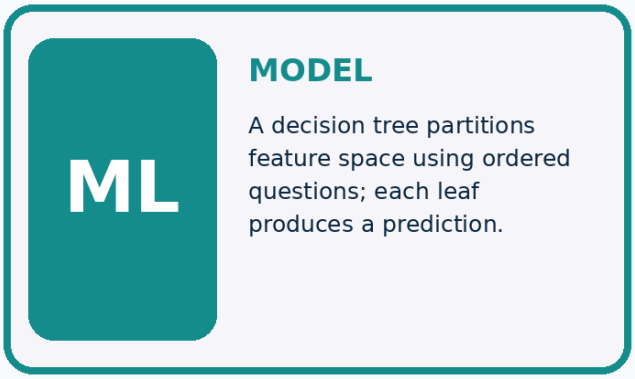
> Follow the mechanism for Decision Trees. Connect each labelled input, deciding step and result.

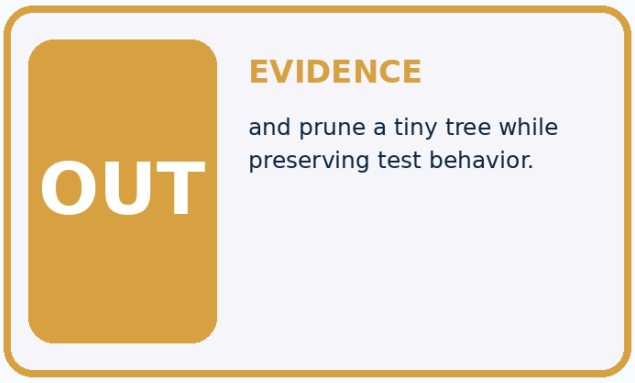
ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Trace, draw, and prune a tiny tree while preserving test behavior.

MODEL IDEA A decision tree partitions feature space using ordered questions; each **leaf** produces a **prediction**.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD **Root** asks whether spot_count ≥ 3. Yes **branch** predicts spotted; no **branch** asks whether yellow_ratio ≥ 0.25.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

```
rows = [(1, 0, "blocked"), (1, 1, "blocked"),
        (1, 0, "blocked"), (0, 1, "clear"),
        (0, 0, "clear"), (0, 1, "blocked")]
def majority(labels):
    if labels.count("blocked") > labels.count("clear"):
        return "blocked"
    return "clear"
best_errors = len(rows) + 1
```

```
for feature in [0, 1]:
    leaves = {}
    for value in [0, 1]:
        labels = [row[2] for row in rows
                  if row[feature] == value]
        leaves[value] = majority(labels)
    errors = 0
    for row in rows:
        if leaves[row[feature]] != row[2]:
            errors = errors + 1
    print(feature, errors)
    if errors < best_errors:
        best_errors = errors
        tree = (feature, leaves)
print(tree[0], tree[1][0], tree[1][1])
```

Feature 0 is near; feature 1 is tall; row[2] is the label. **Output** is 0 1, then 1 2, then 0 clear blocked. Near makes one **training** mistake: near=0 contains two clear labels and one blocked, so that **leaf** predicts clear.

Near=1 contains three blocked labels. Tall makes two mistakes because both its groups have a blocked majority.

The stored tree contains the chosen feature and its two learned leaves. The code assumes nonempty data, binary flags, allowed labels and examples on both **branch**es of each candidate.

An empty **branch** would need a declared fallback; this small example does not implement one. The score is total **training** mistakes, not an average giving small and large **branch**es equal weight.

This is an explicitly limited **depth**-one learner. General tree algorithms can repeat **split** selection within **branch**es and often use impurity criteria such as Gini. Our code does not implement that full procedure.

Keeping the scope small lets you inspect every decision.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

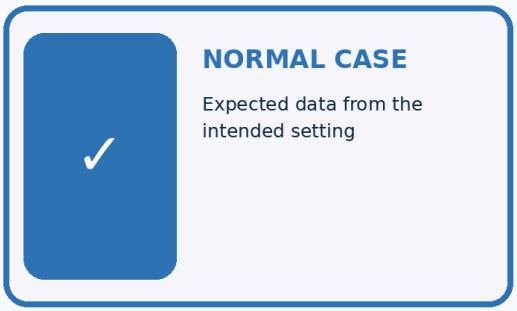
Compare the cases

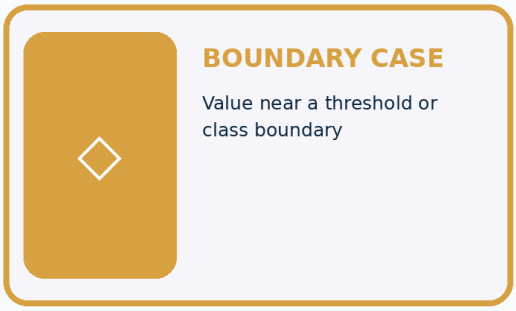
> Compare the supplied cases for Decision Trees. Identify the detail that changes the conclusion.

EVALUATION DESIGN

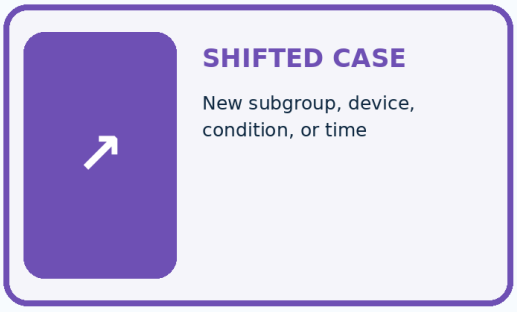
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a **threshold**, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: The stored rule changes to visible_spots>2.

Predicted result

Prediction: M is now predicted healthy.

Why the result follows

Reason: Changing >= to > moves the exact value 2 to the other **branch**.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Decision Trees to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.

1

SPLIT

question dividing examples

2

THRESHOLD

numeric boundary used in a split

3

LEAF NODE

terminal prediction

4

PRUNING

remove unnecessary branches

Term	Precise meaning	My ML example
split	question dividing examples	
threshold	numeric boundary used in a split	
leaf node	terminal prediction	
pruning	remove unnecessary branches	

MISCONCEPTION TO REPAIR A deeper tree must be better because it asks more detailed questions.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Root	The first question asked by a tree
Branch	A path selected by the answer to a question
Leaf	An endpoint that stores a prediction
Depth	The largest number of question steps from root to leaf

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Decision Trees, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 11 • WORKED EVIDENCE

1. Root asks whether spot_count ≥ 3.
2. Yes branch predicts spotted; no branch asks whether yellow_ratio ≥ 0.25.
3. Trace leaf A: spot_count=1, yellow_ratio=0.10 → healthy.
4. A branch fitting one training oddity hurts test results, so prune and retest.

Step 1

STEP 1 **Root** asks whether spot_count ≥ 3.

Step 2

STEP 2 Yes **branch** predicts spotted; no **branch** asks whether yellow_ratio ≥ 0.25.

Step 3

STEP 3 Trace **leaf** A: spot_count=1, yellow_ratio=0.10 → healthy.

Step 4

STEP 4 A **branch fitting** one **training** oddity hurts test results, so prune and retest.

CHECK Verify the denominator, unit, **split**, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Two paths through the learned tree

ROOT QUESTION	BRANCH ANSWER	LEARNED LEAF
Is near equal to 1	No near is 0	Predict clear
Is near equal to 1	Yes near is 1	Predict blocked

Both paths start at the same root. Tall is not used by this fitted tree; the leaf labels came from training counts.

Worked explanation · Decision Trees

Validation rows are (near=1,tall=1,blocked), (0,0,clear), (0,1,blocked), (1,0,clear). Follow only near: **prediction**s are blocked, clear, clear, blocked. Two of four match.

The **training**-majority baseline predicts blocked, since **training** has four blocked and two clear; it also matches two of four validation labels.

Which validation rows does the tree miss, and which does the baseline miss? Explain why the tall value cannot change this tree's prediction.

Keep the fitted **root** and leaves unchanged while scoring. Equal accuracy does not mean identical errors, and five correct **training prediction**s out of six is not the validation result.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Decision Trees

Trace a small fitted tree, including its exact boundary.

The facts we will use

Supplied facts

Toy stored tree: if visible_spots>=2 predict spotted; otherwise healthy. Missing visible_spots requires review. **Input**s L=1, M=2, N=None.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/11

Decision Trees

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read first branch

L: 1>=2 is false, so predict healthy.



2. Read boundary

M: 2>=2 is true, so predict spotted.



3. Stop missing path

N lacks a count, so use review rather than inventing zero.

CHANGE ONE THING

The stored rule changes to visible_spots>2.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read first branch	L: 1>=2 is false, so predict healthy.
2. Read boundary	M: 2>=2 is true, so predict spotted.
3. Stop missing path	N lacks a count, so use review rather than inventing zero.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. N lacks a count, so use review rather than inventing zero. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Applying a stored tree is inference. Selecting a **threshold** from labelled **training** data is a separate **fitting** step.

Keep the stated limit

Read the labelled picture: Which **input** distinguishes the original and changed tree? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: The stored rule changes to visible_spots>2. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Decision Trees claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: A deeper tree must be better because it asks more detailed questions. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE A deeper tree must be better because it asks more detailed questions.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Record the two candidate questions, **branch** label counts, total errors, tie rules, **depth** limit, stored tree and validation **prediction**s.

Retain all rows, including the blocked example in the near=0 **branch**; removing a disagreement would change the **evidence**.

Check the claim against the facts

Our tree uses a single feature because that question scored better under this procedure. This neither establishes that near causes blockage nor that tall is useless in every dataset.

Next week a nearest-neighbour method will predict by comparing examples instead of following learned questions.

A learner says the tree is reliable because near=1 has only blocked training labels. Rewrite this claim using the leaf evidence, validation score and one limitation.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Decision Trees with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

Question to explain

QUESTION + intended use + success **threshold**

DATA version + grouped **split** + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Do not assume near=1 must always predict blocked. If a new **training** set gives that **branch** a clear majority, this same learner will store clear there. A feature name is not the answer.

Use the **training** labels belonging to that **branch**.

Adding more questions can fit finer **training** details but does not guarantee better new-case performance. Record any **depth** choice made during development, and keep final test data separate.

A **leaf** majority describes this **training** group; it is not a promise about every future example.

Practise with less help: Verify the candidate scores

Use workbook W2 for the connected example "**Fitting** a decision tree". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Fitting** a decision tree. Proposed explanation: "A near **branch** must predict blocked". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which **input** distinguishes the original and changed tree? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

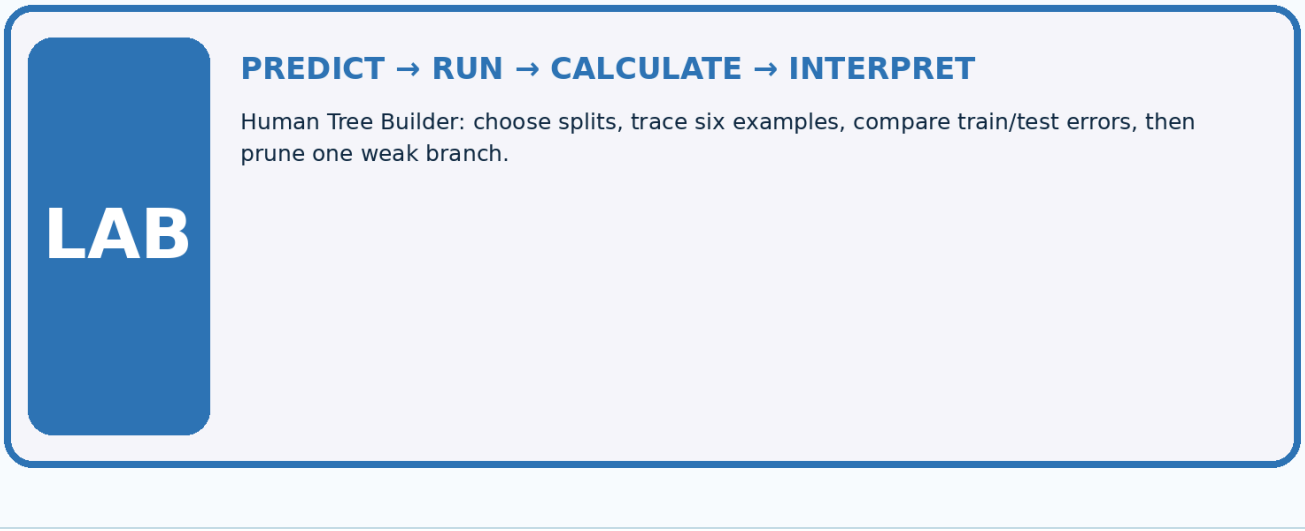
Run the investigation

> Investigate Decision Trees. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Human Tree Builder: choose **split**s, trace six examples, compare train/test errors, then

prune one weak **branch**.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal
Prediction: _____
Actual: _____
Calculation / evidence: _____
Meaning / next test: _____

Case: Boundary
Prediction: _____
Actual: _____
Calculation / evidence: _____
Meaning / next test: _____

Case: Missing
Prediction: _____
Actual: _____
Calculation / evidence: _____

> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

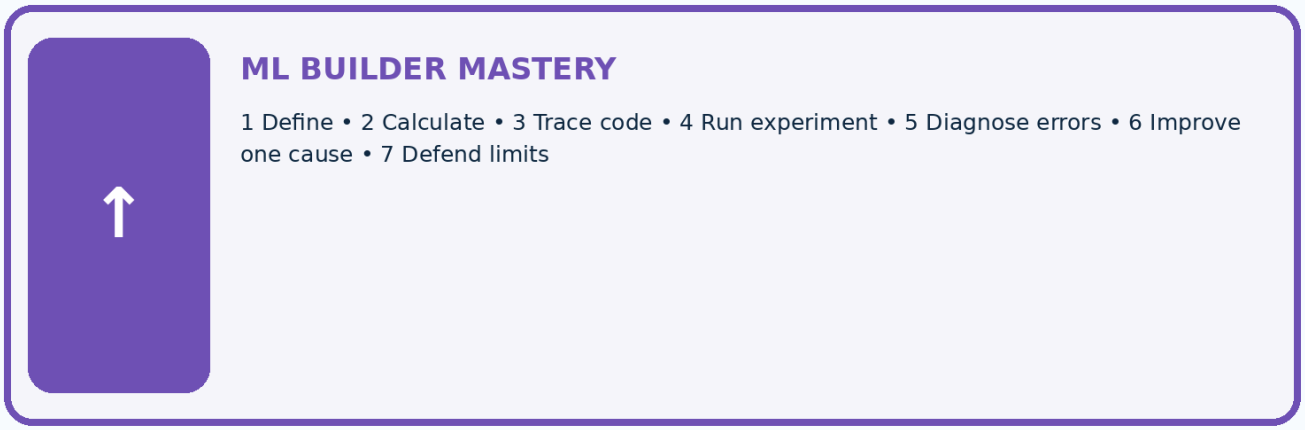
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Decision Trees. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **split**, **threshold**, **leaf node**, **pruning**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE A deeper tree must be better because it asks more detailed questions.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is

completed independently.

STUDENT LESSON · 11 OF 18

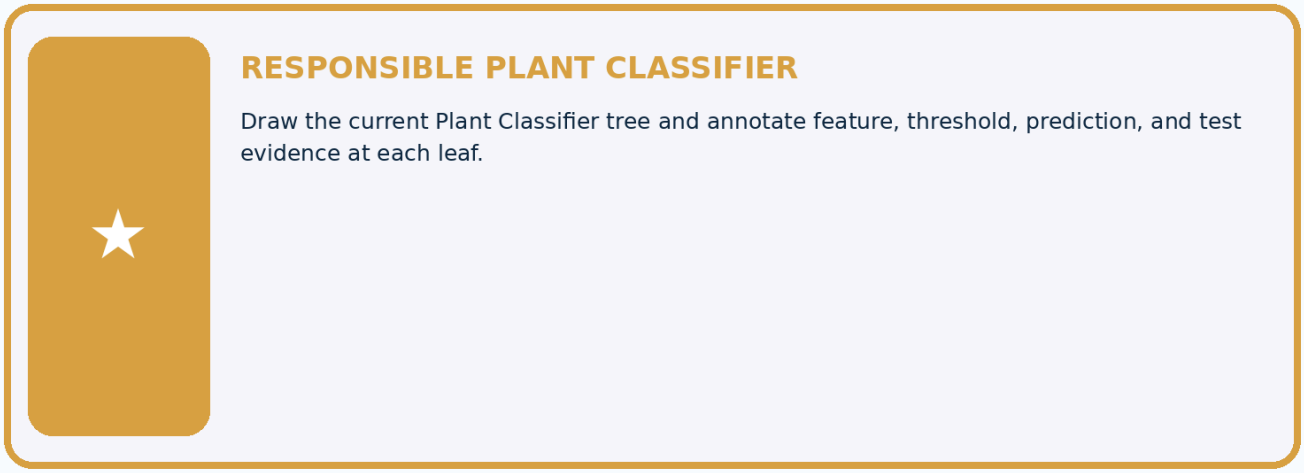
Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



THIS WEEK'S CHECKPOINT Draw the current Plant Classifier tree and annotate feature, **threshold**, **prediction**, and test **evidence** at each **leaf**.

Which requirement or risk does this artifact address?

Which dataset, **split**, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

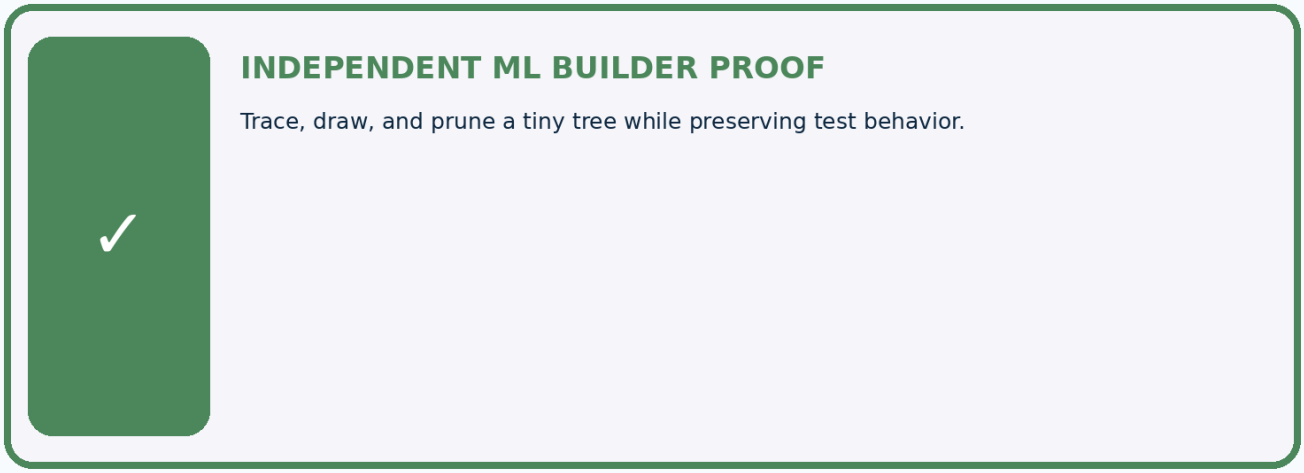
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Trace, draw, and prune a tiny tree while preserving test behavior.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

> **Connect to the next lesson**
>
> NEXT BRIDGE Carry the experiment **evidence** into: Nearest Neighbors.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Decision Trees case for supported practice. Repeated examples do not become fresh assessments.

Week 11 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A supplied tree predicts spotted for spot_count>=3, otherwise healthy.

The question

Trace 2,3,4 and state boundary.

Worked explanation

Healthy, spotted, spotted. Three is included. This supplied rule does not show an optimal learned **threshold**.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Trace the inclusive tree threshold

Evidence or stage	Value or result
2 spots	Below 3 → healthy
3 spots	At least 3 → spotted
4 spots	At least 3 → spotted

A supplied tree predicts spotted for spot_count>=3, otherwise healthy.

Trace 2,3,4 and state boundary.

Healthy, spotted, spotted. Three is included. This supplied rule does not show an optimal learned **threshold**.

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Changed independent case

A stored fitted rule says spots >=5 predicts spotted; otherwise healthy. Trace4,5,6 and a missing count. Policy: missing count requires review. Can test labels be used to choose this cutoff?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 11 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A tree predicts fern when **leaf**_width_cm <= 3 and grass otherwise. Trace widths 2, 3 and 4. Which case checks the boundary?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Draw the current Plant Classifier tree and annotate feature, **threshold**, **prediction**, and test **evidence** at each **leaf**.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

> Rule: if spots>=4 predict spotted, otherwise healthy. Trace 3,4,5 and missing. Does applying this rule demonstrate that **training** found the **threshold**?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Decision Trees, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Trace a small fitted tree, including its exact boundary.

By the end, I can	How I will show it
Fit a root question and leaf predictions from training examples	Explain the deciding step, show a trace or test, and state one limit.
Trace a new example through the learned tree	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Compare a shallow tree with a baseline and state its limits	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Decision Trees without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Fit a **root** question and **leaf prediction**s from **training** examples

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Trace a small fitted tree, including its exact boundary.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? N lacks a count, so use review rather than inventing zero.

1. Read boundary

2. Read first **branch**

3. Stop missing path

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage

that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Stop missing path N lacks a count, so use review rather than inventing zero. Read the picture in order. N lacks a count, so use review rather than inventing zero.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The stored rule changes to visible_spots>2. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. M is now predicted healthy.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: M is now predicted healthy. Changing >= to > moves the exact value 2 to the other **branch**.

Repair a missing building block

Review Baselines First (ai-4/10). Goal: Fit a majority label baseline using **training** labels only. Short practice: Can the candidate's **training** accuracy replace this validation comparison?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Leakage (ai-4/8). Goal: Choose a **split** unit that matches the intended **prediction** claim. Short practice: How many independently new plant identities are in the original test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Inspect Before Modeling (ai-4/5). Goal: Inspect row meaning missing values and repeated records. Short practice: Which original information should the inspection log preserve?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

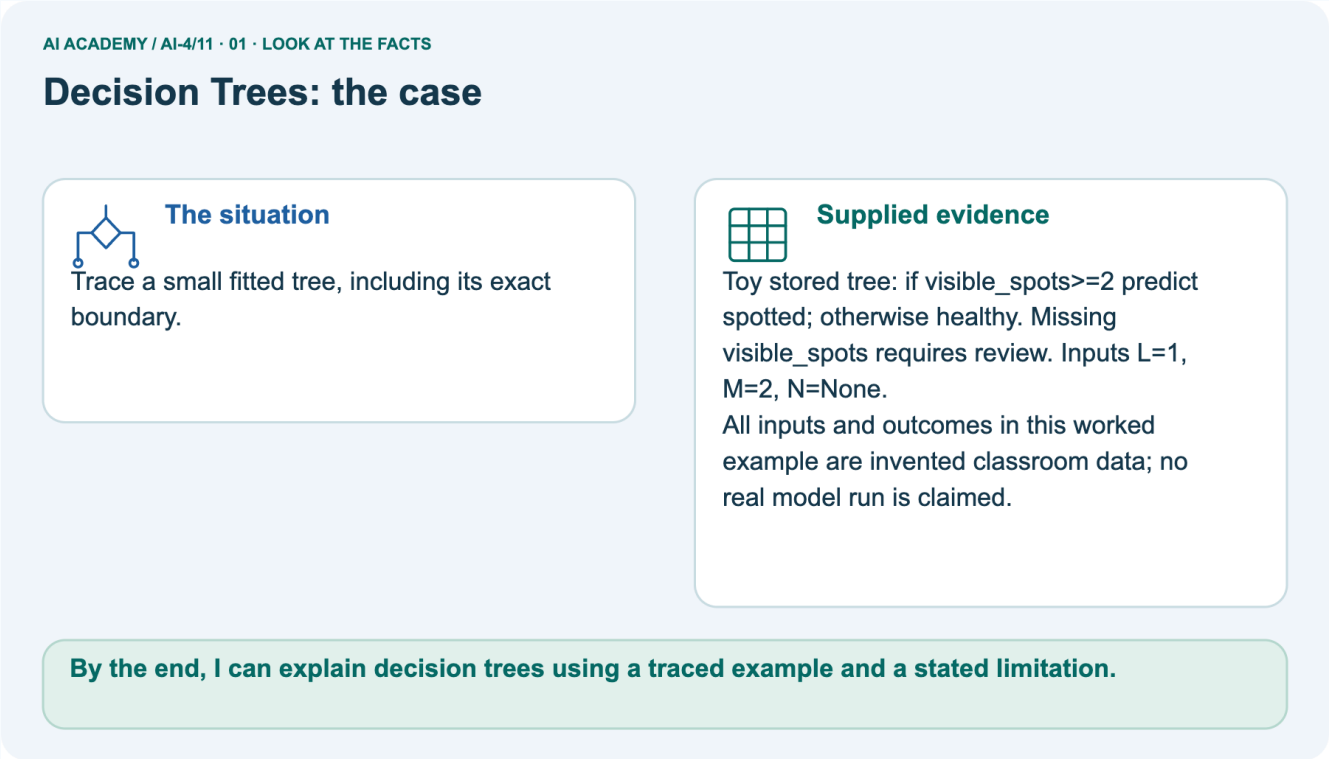
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Decision Trees graphic using its labels, deciding evidence and relevant limitation.

01. Decision Trees: the case



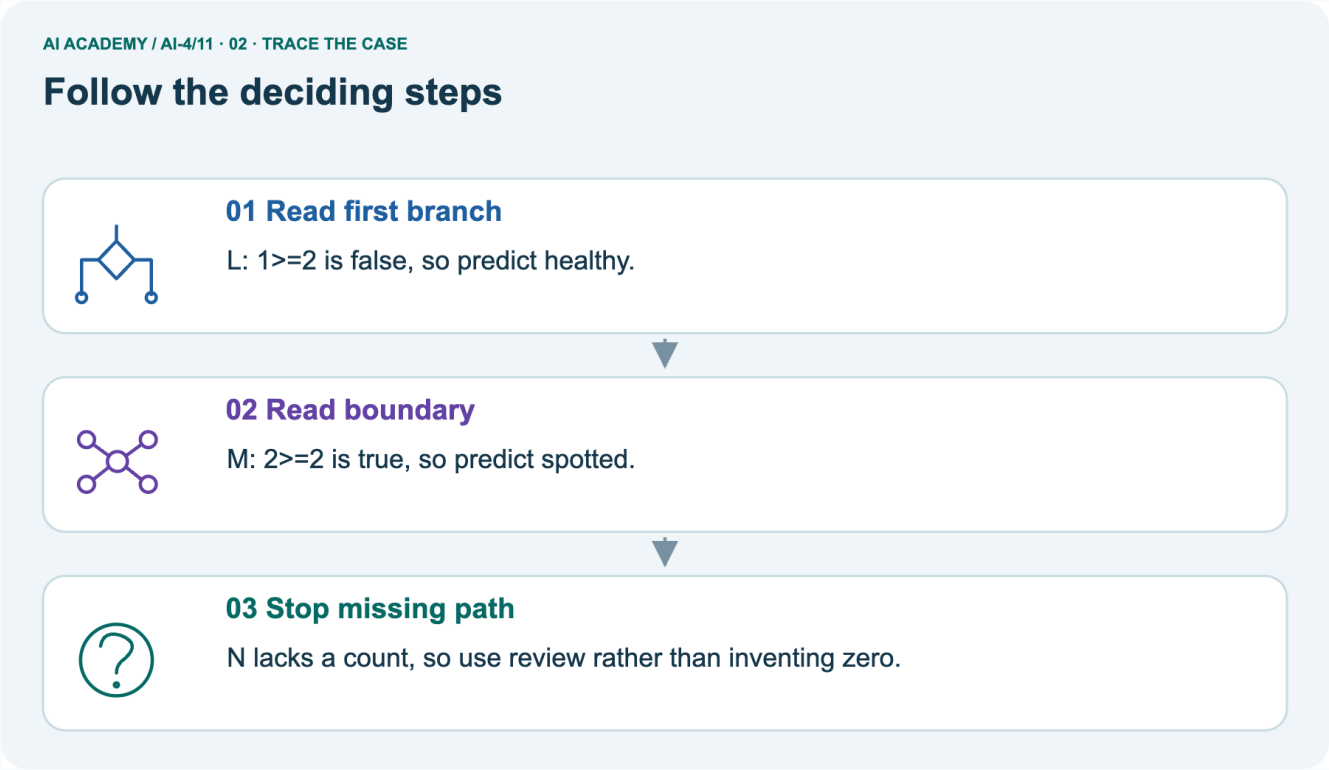
Follow the labelled picture with the matching explanation. Open the vector diagram

Trace a small fitted tree, including its exact boundary. Supplied facts: Toy stored tree: if visible_spots>=2 predict spotted; otherwise healthy. Missing visible_spots requires review. **Input**s L=1, M=2, N=None.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read first **branch**: L: 1>=2 is false, so predict healthy. → Read boundary: M: 2>=2 is true, so predict spotted. → Stop missing path: N lacks a count, so use review rather than inventing zero.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Two paths through the learned tree

AI ACADEMY / AI-4/11 · 03 · INSPECT THE LESSON MODEL

Two paths through the learned tree

Root question	Branch answer	Learned leaf
Is near equal to 1	No near is 0	Predict clear
Is near equal to 1	Yes near is 1	Predict blocked

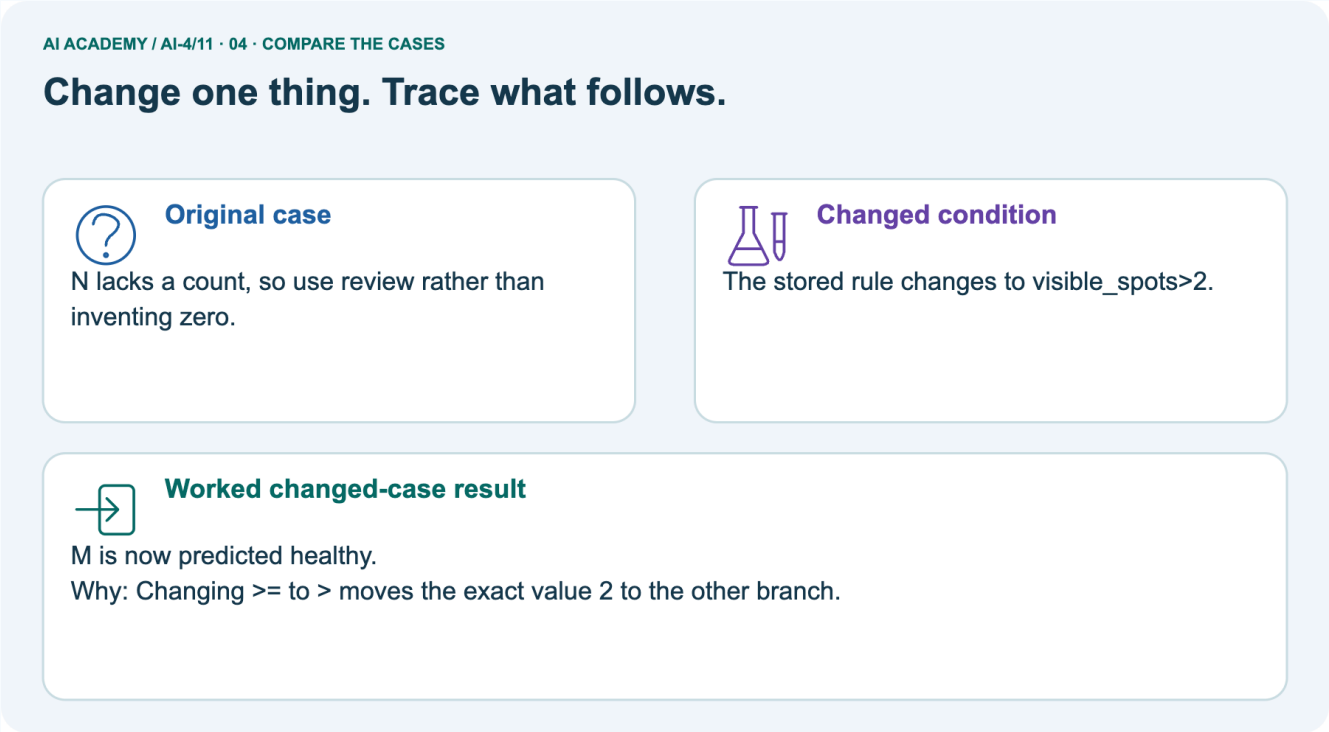
Both paths start at the same root. Tall is not used by this fitted tree; the leaf labels came from training counts.

Follow the labelled picture with the matching explanation. Open the vector diagram

Two paths through the learned tree. Is near equal to 1 | No near is 0 | Predict clear. Is near equal to 1 | Yes near is 1 | Predict blocked

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



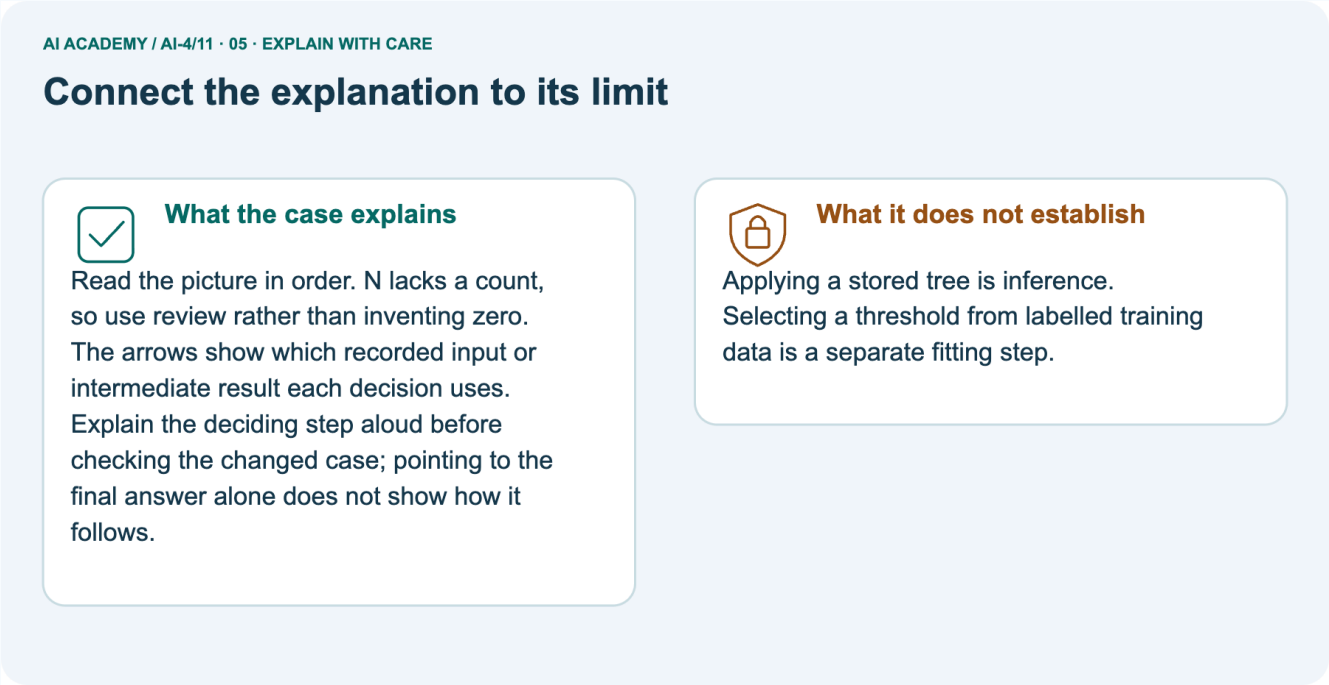
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: N lacks a count, so use review rather than inventing zero. Changed condition: The stored rule changes to visible_spots>2. Expected result: M is now predicted healthy.

Reason: Changing >= to > moves the exact value 2 to the other **branch**.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

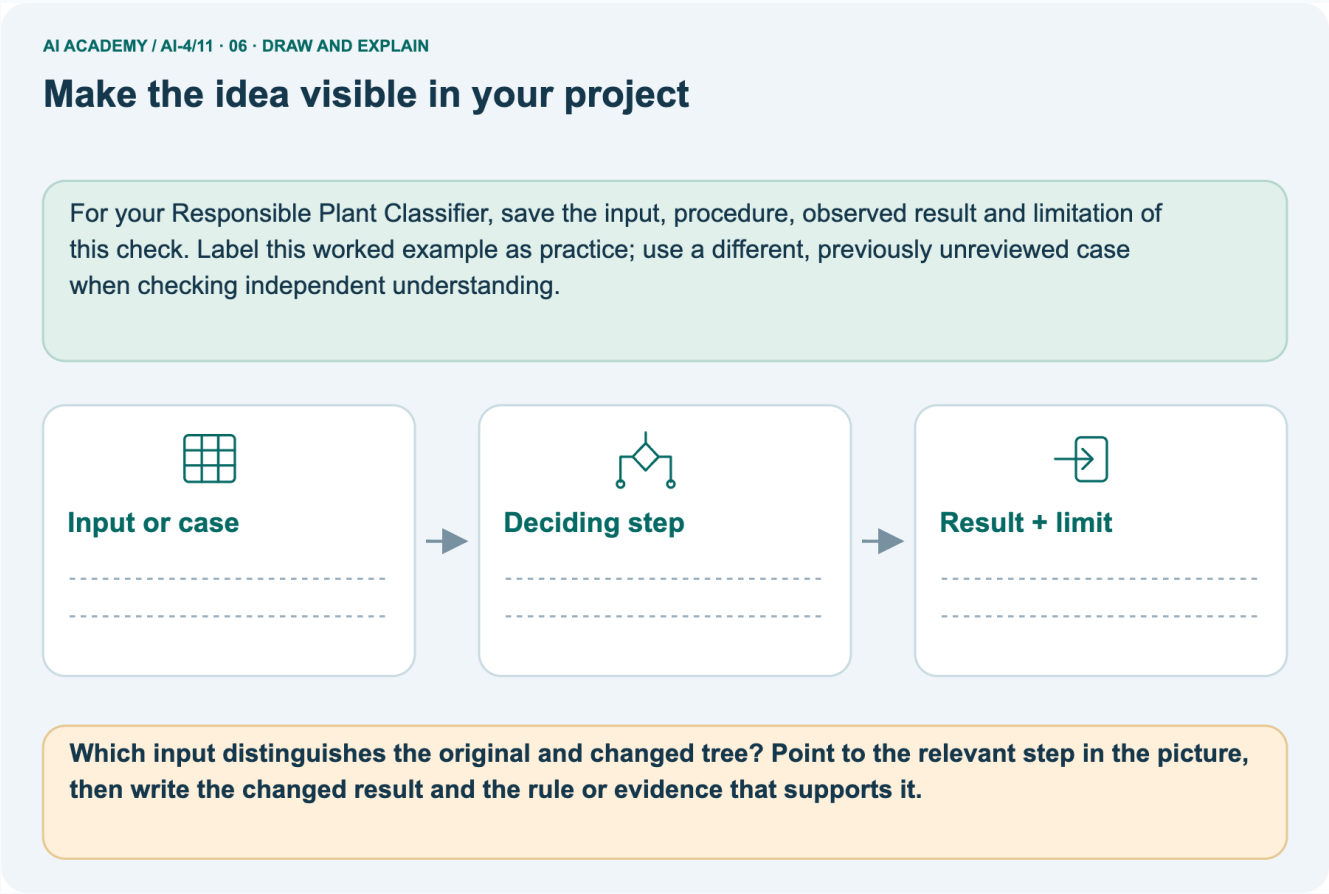
Read the picture in order. N lacks a count, so use review rather than inventing zero. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Applying a stored tree is inference.

Selecting a **threshold** from labelled **training** data is a separate **fitting** step.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which **input** distinguishes the original and changed tree? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.